

Chapter 8 One-Pager — Safe Exploitation

Research on the possibilities for applying Artificial Intelligence in computer games

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Problem. Chapter 7 built the *sensor* — an opponent model — and showed its danger: best-responding to an imperfect model can open a leak in your own play (the continuous model *lost* to Nash on Leduc). Chapter 8 builds the *actuator*: turn a model into profit **without becoming exploitable**. This is the second half of the thesis’s Behavioral Adaptation Framework (Contribution #1) and the launch point for multi-agent safe exploitation (Contribution #2).

Approach. Every safe-exploitation method is the same program — *maximize expected value against the opponent model, subject to a safety floor on your worst-case value* — which is linear in the hero’s sequence-form realization plan. One LP engine solves it by **constraint generation** (solve → read policy → call an exact best response as the worst-case oracle → add a cut if unsafe → re-solve), reusing Chapter 7’s engines, best-response, Nash and opponent zoo. Five families differ only in the floor: **RNR** (tunable p), **best equilibrium** (Ganzfried–Sandholm’s baseline; $\geq$ Nash value v^*), **prime-safe** ($\geq v^* - \epsilon$ for an ϵ -equilibrium baseline), **SES** (blueprint value, enforced locally on a subgame via a gadget), **adaptation** (no more exploitable than the blueprint). Testbeds: Kuhn and Leduc, both exactly solvable, so every result is bracketed by exact analytical references. **All numbers below are measured.**

Key results (measured).

- *Safe exploitation works on Kuhn, but gains little. Best equilibrium is safe against every opponent* (worst-case $\geq v^*$ within $1e-3$) **and beats Nash’s own EV on every exploitable type** — e.g. **+0.222 vs Nash’s +0.146** against **AlwaysPass** — but only by 0.002–0.076 per hand (1–9 % of the full best response’s gain): a per-hand floor at v^* admits only equilibrium strategies. Full best response earns more (+0.975) but its worst-case collapses to **−0.5** (ruinously exploitable).
- ϵ *is measured, not fabricated*. Prime-safe/adaptation lower the floor to $v^* - \epsilon$ with the baseline’s exploitability measured at $\epsilon = \mathbf{0.0074}$, and earn a little more than best equilibrium by spending that budget (+0.266 vs +0.222 on **AlwaysPass**).
- *Canonical RNR switches bang-bang, not along a smooth curve* (a contradicted prediction). In a game this small the max-min LP jumps from the safe vertex straight to full best response between $p = 0.6$ and 0.7 (sampled in steps of 0.1); mixing the two vertices traces the chord, which beats the naive blend by only ≈ 0.002 here.
- *Headline — within a 40-iteration cap the global solvers did not converge on Leduc; the subgame method (cap 400) did*. On **Leduc**, the global solvers (best equilibrium/prime-safe/adaptation) **fail to converge within a 40-iteration cap** (worst-case **−0.64 to −1.33**, grossly unsafe), while the **subgame method (SES) converges** (194–350 iters) and stays near-safe (worst-case $\approx \mathbf{−0.13}$) while extracting **+0.25 to +0.68** vs weak types. This is the global-vs-local safety gap, measured on a tiny game — but at unequal budgets, so it shows the global loop is slow, not that it cannot converge. Chapter 14’s one-shot dual LP solves full Leduc in 0.02–0.04 s.
- *The safety guarantee bites against a worst-case adversary, not a benign one*. In the teaching attack (bait → Nash reveal), the honest separating signal is the safety-violation count (**full_br 40/40 refits, best equilibrium 0/40**), not realized profit — a gentle Nash “revealer” never claws back full_br’s bait-phase windfall. A punishing test needs an adaptive counter-exploiter.

Thesis connection. Chapter 7 = sensor; Chapter 8 = actuator. The Kuhn results show the actuator behaves as the theory predicts (perfect models, an exactly solvable game); the Leduc non-convergence is the concrete argument for real-time subgame methods (SES / OX-Search) and/or an exact dual-LP formulation, and it is the empirical bridge into the scalable, multi-agent safe exploitation of Contribution #2. Chapter 15 adds the stronger two-player baseline, Ganzfried–Sandholm’s RWYWE: +0.062 (Kuhn) and +0.113 (Leduc) chips per hand over the blueprint against best equilibrium’s +0.020 and +0.047, never below the game value in 120 matches under teaching attacks, yet only 14–30 % of the gain of RNR at $p = 0.5$: few gifts are provable when cards are seen only at showdown.

Open questions. Scalable safety beyond Leduc (exact dual-LP vs local/subgame); whether SES is provably safe (its 0.04 residual is measured against v^* ; it holds its own blueprint floor only to the 0.01 tolerance); a punishing teaching attack (adaptive reveal); and **N-player safety** — every guarantee here rests on the two-player zero-sum fact that Nash secures v^* against any opponent, an anchor that vanishes for $N > 2$; the N-player notions that exist are very conservative (team maxmin) or not securable in general (equal share), so exploitation with a baseline-relative loss bound, also against colluding opponents, is open (Contribution #2).